Modern Academy
for Engineering and Technology
Basic Science Department

Academic Year: 2017-2018
Semester: Summer
Exam Date: 14/8/2018

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (MTH203) Differential Equations and Transforms **Spec:** Man. Eng. & Prod. Tech.
Examiners: Assoc. Prof. Dr. Ashraf Taha and Prof. Dr. Aly Essawi **Time:** 3 hours

Number of Pages: 2 Number of Questions: 5 Attempt all questions

Question 1 (14 points)

Solve the following differential equations:

a) $\frac{dy}{dx} = e^{-y} + x^2 e^{-y}.$ (2 points)

b) $\frac{dy}{dx} = \frac{y^2}{x^2} + \frac{5y}{x} + 4.$ (4 points)

c) $\left(y^2 + x^2\right)dx + \left(2xy\right)dy = 0.$ (4 points)

d) $\frac{dy}{dx} - \frac{1}{2x}y = 5x^2y^5.$ (4 points)

Question 2 (16 points)

Find the general solution of the following differential equations:

a) $\frac{d^3y}{dx^3} - 4\frac{d^2y}{dx^2} + 5\frac{dy}{dx} - 2y = 0.$ (4 points)

b) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 10y = 2e^{3x}.$ (4 points)

c) $\frac{d^2y}{dx^2} + y = \tan(x).$ (4 points)

d) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^x,$ given that $y = e^{2x}$ is a homogenous solution. (4 points)

Question 3 (16 points)

a) ***Solve the following differential equation using Laplace transform:***

$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 8y = e^{3t}; \qquad y(0) = 0, y'(0) = 2.$ (4 points)

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b) ***Find the Laplace transform of the following functions:***

i. $f(t) = t^5 + 7e^{6t} + 8\cosh 2t.$ (3 points)

ii. $f(t) = \int_0^t e^{\tau} \sin 4\tau \ d\tau.$ (3 points)

iii. $f(t) = u_3(t) \sin^2(t-3).$ (3 points)

iv. $f(t) = \dfrac{\sin t}{t}.$ (3 points)

Question 4 (16 points)

a) ***Find the inverse Laplace transform of the following functions:***

i. $F(s) = \ln\left(\dfrac{s^2+16}{(s-1)^2}\right).$ (4 points)

ii. $F(s) = \dfrac{s}{s^2+10s+8}.$ (4 points)

iii. $F(s) = e^{-4s}\left(\dfrac{3s}{s^2+11}\right).$ (4 points)

b) ***Represent the following function*** $f(x) = x^2, \ 0 \le x \le \pi$ ***as Fourier Sine Series*** (4 points)

Question 5 (8 points)

a) ***Prove the recurrence relations*** $xP_n(x) = \dfrac{n+1}{2n+1}P_{n+1}(x) + \dfrac{n}{2n+1}P_{n-1}(x).$ (4 points)

b) ***Evaluate the integral*** $\int_{-1}^{1} xP_1(x) \ dx.$ (2 points)

c) ***Evaluate the integral*** $\int_{-1}^{1} x\,P_2(x)P_3(x) \ dx.$ (2 points)

GOOD LUCK